



Name: _____

Teacher: _____

SCEGGS Darlinghurst

HSC Assessment 1

14 February, 2003

Mathematics – Extension 1

General Instructions

- Time allowed: 60 minutes
- Weighting 25%
- This paper has **three** questions
- Attempt **all** questions and show all necessary working
- Marks will be deducted for careless or badly arranged work
- Write using blue or black pen, diagrams in pencil
- Start each question on a new page
- Write your name on each page
- Approved calculators, mathematical templates and geometrical instruments may be used

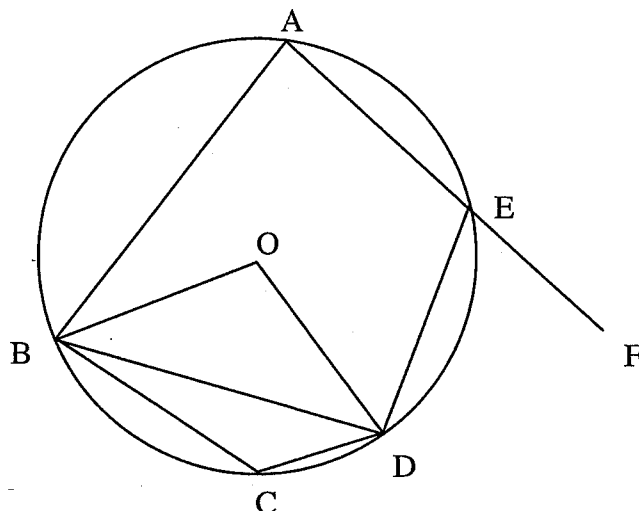
Question		Marks	Communication	Reasoning
1		/13	/5	/4
2		/11	/2	/4
3		/10	/2	/5
	TOTAL	/34	/9	/13

Question One (13 marks)

- Answer on the pad paper provided
- Write your name and at the top of the page
- Start each question on a new page
- Clearly label each question

Marks

(a)



Given: O is the centre of the circle
 $\angle BCD = 150^\circ$
 $\angle ABO = 40^\circ$

- (i) mark all relevant information on the diagram (no reasons necessary)
- (ii) state the value of $\angle FED$

③ C

1

(b) The polynomial:

$$P(x) = x^3 + 2x^2 - x + K$$

is divisible by $x - 1$

- (i) Find the value of K
- (ii) Factorise $P(x)$ fully.
- (iii) Solve $P(x) \leq 0$

1

2

② C

(c) Prove, using Mathematical Induction, that $13 \times 6^n + 2$ is divisible by 5.
 (n Positive integer.)

④ R

Question Two (11 marks)

• START A NEW PAGE

Marks

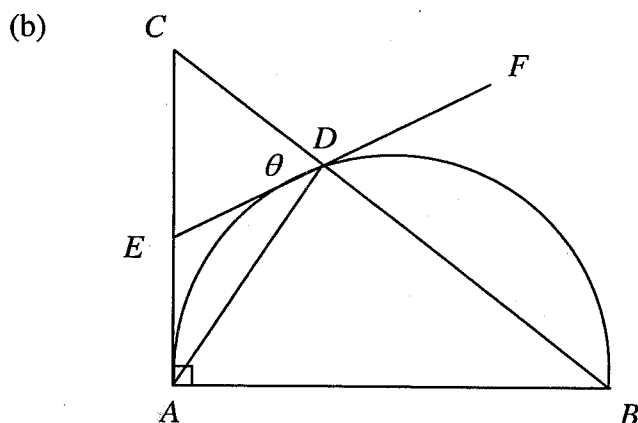
- (a) The roots of the equation:

4

$$x^3 - 12x^2 + 12x + 80 = 0$$

are 3 consecutive terms in arithmetic progression.

Find these roots.



AB is the diameter of a semi-circle. $\triangle ABC$ is right-angled at A , and BC cuts the semi-circle at D . EF is a tangent to the semi-circle at D . AD is joined. $\angle CDE = \theta$.

- COPY OR TRACE THE DIAGRAM ONTO YOUR WRITING PAGE.

- | | |
|---|-----|
| (i) Why is $\angle ADB = 90^\circ$? | ① c |
| (ii) Why is $\angle ADE = \angle ABD$? | ① c |
| (iii) Name two angles equal to $\angle CDE$. | 1 |
| (iv) Prove $\triangle ADE$ is isosceles. | ② R |
| (v) Prove that E is the midpoint of AC . | ② R |

Question Three (10 marks)

• **START A NEW PAGE**

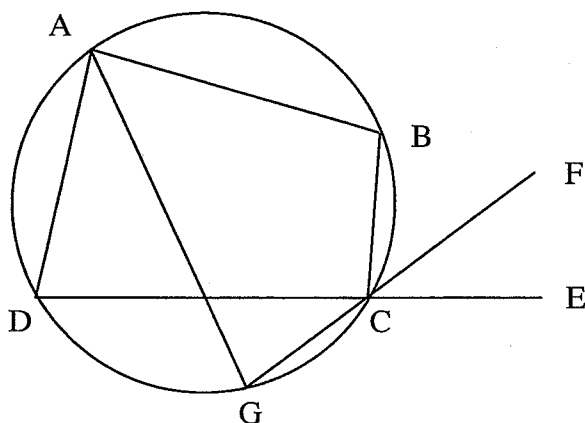
Marks

- (a) Use Mathematical Induction to prove that for all positive integers n

⑤ R

$$\sum_{r=1}^n r(r!) = (n+1)! - 1$$

- (b)



Given AG bisects $\angle DAB$,

Prove that:

(i) $\angle DAG = \angle FCE$.

② C

(ii) $\angle DAB = \angle BCE$.

1

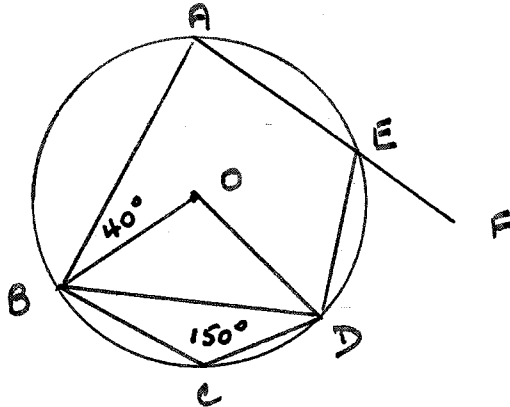
(iii) hence CF bisects $\angle BCE$.

2

End of Paper

Extension 1 Assessment Test 1 14/2/03

① a) (i)



✓ radii equal ($BO = OD$)

✓ reflex $\angle BOD = 300^\circ$
(or $\angle BOD = 60^\circ$)

✓ $\angle ABD = \angle FED$

✓ (ii) $\angle FED = 100^\circ$

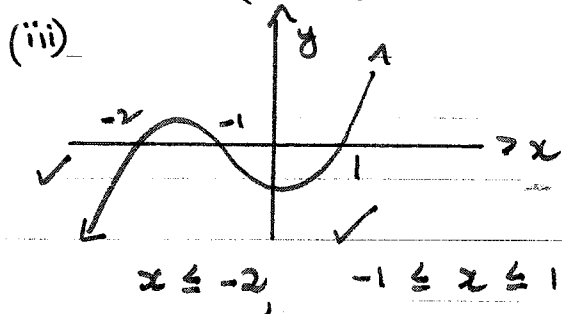
b) (i) $P(1) = 1 + 2 - 1 + 2 = 0$

✓ $12 = -2$

(ii) ✓

$$\begin{array}{r} x^2 + 3x + 2 \\ x-1 \overline{) x^3 + 2x^2 - x - 2} \\ \underline{x^3 - x^2} \\ 3x^2 - x - 2 \\ \underline{3x^2 - 3x} \\ 2x - 2 \end{array}$$

✓ $P(x) = (x-1)(x+2)(x+1)$



c) If $n=1$, $13 \times 6 + 2 = 80$
which is divisible by 5
Assume true for $n=k$
i.e. $13 \times 6^k + 2 = 5M$
(M an integer)

consider $n = k+1$

$$\begin{aligned} 13 \times 6^{k+1} + 2 &= 13 \times 6^k \times 6 + 2 \\ &= (5M - 2) \times 6 + 2 \\ &= 30M - 12 + 2 \\ &= 30M - 10 \end{aligned}$$

which is divisible by 5.

if true for $n=k$, the statement is true for $n=k+1$. It is true for $n=1$ and thus is true for $n=2, 3$ etc.

i.e. true for all n positive integer.

② a) If in A.P. let the roots be $a-d, a, a+d$. ✓

$$\therefore a-d + a + a+d = 12$$

$$3a = 12$$

$$a = 4$$

(i) ✓

$$(a-d)a(a+d) = -80$$

$$4(16-d^2) = -80$$

$$16-d^2 = -20$$

$$d^2 = 36$$

$$d = 6 \text{ or } -6$$

✓

if $d=6$, terms are $-2, 4, 10$ ✓

if $d=-6$, terms are $10, 4, -2$

$\therefore$ terms are $-2, 4, 10$

b) (i) Angle in a semi circle is 90° ✓
(ii) angle between tangent and chord equals angle in alternate segment ✓
(iii) $\angle FDB, \angle DAB$ ✓

(iv) $\angle CDA = 90^\circ$ (adjacent angles supplementary)

$$\therefore \angle EDA = 90^\circ - \theta$$

$$\angle EAB = 90^\circ \text{ (given)}$$

$$\therefore \angle EAD = 90^\circ - \theta$$

$\therefore \Delta$ is isosceles (2 angles equal)

(v) $EA = ED$ (opp. equal angles in ΔEAD)

$$\begin{aligned} \angle ECD &= 90^\circ - \angle ABD \text{ (angle sum of } \Delta CAB \text{ is } 180^\circ) \\ &= 90^\circ - \angle EDA \end{aligned}$$

$$\text{But } \theta = 90^\circ - \angle EDA \text{ (above)}$$

$$\therefore \angle ECD = \theta$$

$\therefore EC = ED$ (opp equal angles in ΔECD)

$$\therefore EA = EC$$

i.e. E is midpoint AC

$\therefore$ if true for $n = k$, the statement is true for $n = k+1$. It is true for $n = 1$ and is thus true for $n = 2, 3, \dots$ i.e. all n +ve integers

b)

(i) $\angle DAG = \angle DCE$ (standing on same arc, equal)

$\angle FCE = \angle DCE$ (vertically opp. =)

$$\therefore \angle DAG = \angle FCE.$$

(ii) $\angle DAB = \angle BCE$ (ext $\angle$ of cyclic quadrilateral = int. opp. angle)

(iii) $\angle DAB = 2 \angle DAC$ (Arc subtends $\angle DAB$, given)

$$\therefore \angle BCE = 2 \angle DAC$$

but $\angle FCE = \angle DAC$ (proved)

$$\therefore \angle BCF = \angle DAC$$

i.e. CF bisects $\angle BCE$.

③ a) To prove that

$$1.1! + 2.2! + \dots + n.n! = (n+1)! - 1$$

$$\text{If } n=1, \quad 1.1! = 1$$

$$2! - 1 = 1$$

$\therefore$ true for $n=1$

Assume true for $n = k$

$$\text{i.e. } 1.1! + 2.2! + \dots + k.k! = (k+1)! - 1$$

Consider $n = k+1$,

$$\text{L.H.S.} = 1.1! + 2.2! + \dots + k.k! + (k+1)(k+1)!$$

$$= (k+1)! - 1 + (k+1)(k+1)!$$

$$= (k+1)! [1 + k+1] - 1$$

$$= (k+2)(k+1)! - 1$$

$$= (k+2)! - 1$$

$$= (k+1+1)! - 1 = (k+1+1)! - 1$$